phase angle = …… ° [2]

5 **(a)** State the *principle of superposition.*

……..

……..

……..... [2]
[1]

(b) Fig. 5.1 shows the arrangement for viewing a visible interference pattern on a screen.

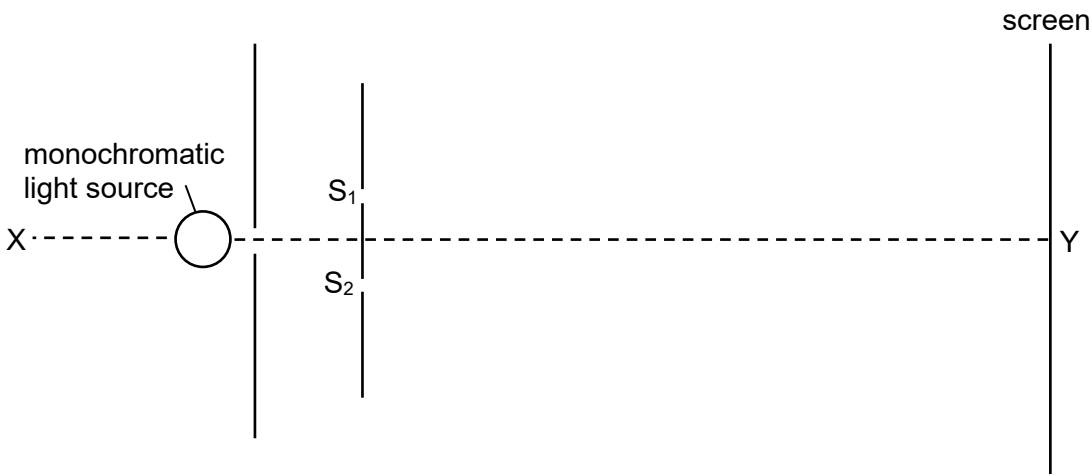

Fig. 5.1

In a darkened room, a double slit (S_1S_2) is placed in front of a narrow single slit situated in front of a monochromatic light source. Line XY is equidistant from S_1 and S_2.

(i) Explain why a single slit is used in Fig. 5.1.

…….....

…….....

…….....

…….....

……..... [3]

(ii) The point Y on the screen is directly opposite to the centre of the double slit. State and explain the nature of the interference that occurs at Y.

…….....

…….....

…….... [2]

(iii) The distance between slits S_1 and S_2 is 0.60 mm. When the screen is placed 1.8 m from the slits, the distance between the two first-order bright fringes in the interference pattern formed on the screen is 4.0 mm. Calculate the wavelength of the light.

wavelength = …… m [2]

(iv) Suggest changes to the appearance of the fringes when each of the following changes is made separately.

1. Moving the light source and single slit along the axis towards point X, further away from S_1S_2.

…….....

…….....

…….... [2]

2. Shifting the single slit a small distance downwards, closer to slit S_2.

…….....

…….....

..... [2]

6 A small coil is positioned so that its axis lies along the axis of a large bar magnet, as shown in Fig. 6.1.